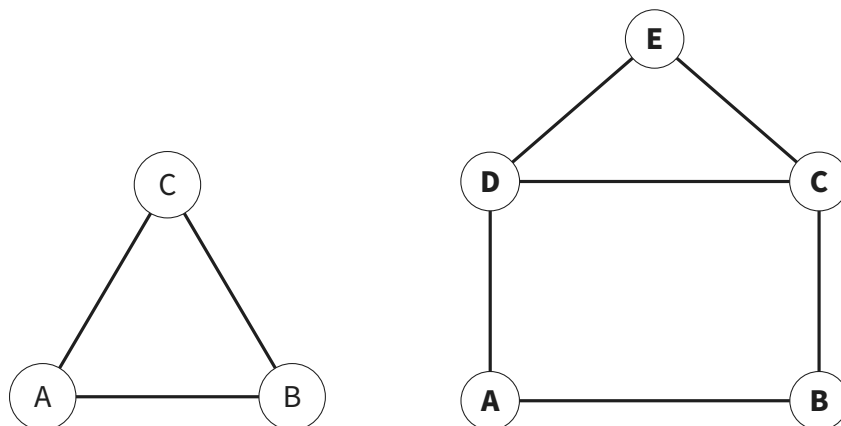


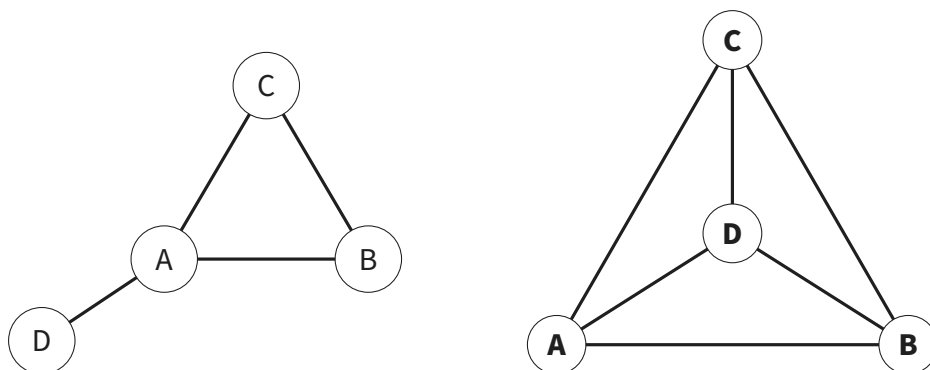
Problem 1: Walk every line once without jumping. You may visit a dot again. Mark each used line with a counter, or number the lines in walking order. Try both maps. You choose the starting dot. Record one route as a list of visited dots, such as A–B–C–A.



Problem 2: Try the house again from A, then from C, then from D. Record a complete route if you find one. Otherwise record the route you tried and the unused lines.

Start	Route	Unused lines
A		
C		
D		

Problem 3: Try to walk each map using every line once. On the right map, D is a junction connected to all three outside dots. Try two different starts on each map. Record where each attempt stops and any unused lines.

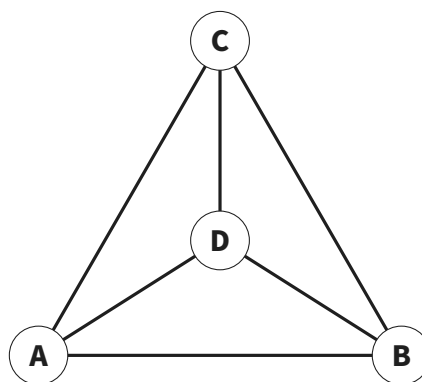
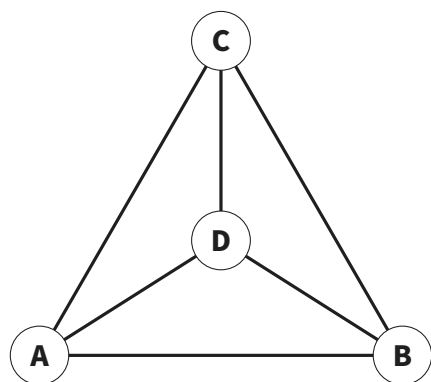


Problem 4: Count the lines meeting at each dot on the house, the triangle with a branch, and the right-hand map above. This count is called degree. Record the counts and circle the odd ones. Compare odd dots with the start and finish of your complete routes.

Map	A	B	C	D	E
House					
Triangle + branch					–
Triangle + center					–

Problem 5: Choose a complete route. At a dot in its middle, pair each arrival with the next departure using two fingers. Record the pairs of lines at that dot.

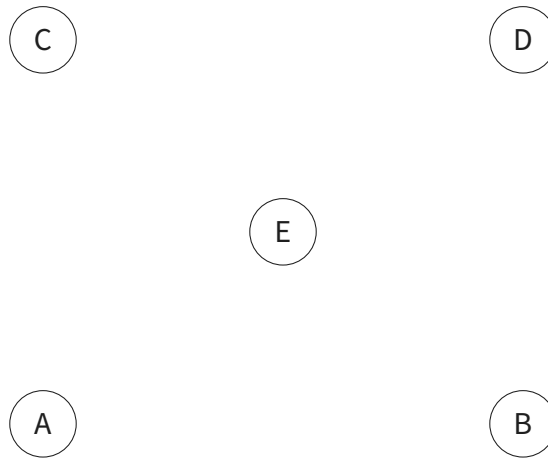
Problem 6: Add a second bridge between A and B on the left map. Draw it curved so the two bridges stay separate. On the right map add a second bridge between A and C. Count the new degrees and circle the odd dots. A crossing without a dot is not a junction.



Problem 7: Find a complete every-bridge-once walk on each repaired map. Start at one of its odd dots. Number the bridges in walking order, keeping both copies separate. Record the two routes.

Problem 8: Choose one repaired map and add a bridge joining its two remaining odd dots. Try to walk every bridge once and return to your start. Record your added bridge and route.

Problem 9: Draw a connected map on the five dots below with exactly two odd dots. Every bridge must join different dots; curved and parallel bridges are allowed. Count each dot's bridges and circle the odd counts.



Problem 10: Find a route using each bridge once. Number its bridges or record its visited dots below. If the attempt gets stuck, try another route or revise a bridge. Keep this solution hidden while a partner tries your map.

Problem 11: Make a second version by adding one bridge. Choose its endpoints so every dot now has even degree. Give your partner the new map and ask for a route that ends where it started. Record the added bridge and a complete route.